Granger Causality

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Introduction

Granger causality tests whether past values of one time series improve the prediction of another, beyond what is already explained by the latter's own history. It is a *statistical* concept, not a *causal* one in the philosophical sense: a series Granger-causes another if including its lags reduces the residual variance of the predicted series, which is a useful but limited notion. Two series can Granger-cause each other without any direct causal link if both are driven by a common unobserved factor.

Prerequisites

A working understanding of vector autoregressive (VAR) models, F-tests on joint coefficient hypotheses, and the difference between predictive and causal claims.

Theory

Series X Granger-causes series Y at lag p if the lagged values $X_{t-1}, \ldots, X_{t-p}$ significantly improve the regression of Y_t on its own lags $Y_{t-1}, \ldots, Y_{t-p}$. The test is an F-test on the joint hypothesis that all coefficients on lagged X are zero. Within a VAR system with K series, Granger causality from one series to another (or to all others) is computed similarly with the residuals jointly estimated.

A non-trivial extension is "instantaneous" causality: X and Y contemporaneously correlated after conditioning on lagged values of both. This is included by some implementations (e.g., `vars::causality`).

Assumptions

Stationary series — the test is invalid for unit-root processes. No unmeasured common cause that drives both series; otherwise the Granger-causal direction is uninterpretable.

R Implementation

```r
library(lmtest); library(vars)

data(ChickEgg)   # eggs / chickens

# Does egg production Granger-cause chicken population?
grangertest(chicken ~ egg, order = 3, data = ChickEgg)
grangertest(egg ~ chicken, order = 3, data = ChickEgg)

# Within a VAR
fit <- VAR(ChickEgg, p = 3)
causality(fit, cause = "egg")$Granger
causality(fit, cause = "chicken")$Granger
```

Output & Results

`grangertest()` returns the F-statistic, degrees of freedom, and p-value for the joint hypothesis of zero coefficients on the cause's lags. `vars::causality()` extends this within a VAR framework, handling the multivariate residual structure correctly and adding the instantaneous-causality test.

Interpretation

A reporting sentence: "Egg production Granger-caused chicken population at lag 3 ($F(3, 48) = 4.9$, $p = 0.005$); the reverse test was not significant ($p = 0.39$). This suggests egg counts contain predictive information about future chicken populations beyond past chicken counts, but does not establish a direct causal mechanism." Always frame Granger results as predictive, not causal.

Practical Tips

- Granger causality requires stationarity; difference or detrend first using KPSS / ADF guidance.
- Results are sensitive to lag order; try several plausible lags and report the chosen value with justification.
- Granger causality does not imply causation; a common confounder produces apparent bidirectional Granger causality between two effects.
- For more than two series, use the VAR-based `vars::causality()` rather than pairwise `grangertest()`; the joint estimation handles cross-series correlations.
- Non-linear Granger causality exists (`bootGran`, `NlinTS`); the standard tests are linear and miss non-linear predictive relationships.
- For frequency-domain Granger causality, the `vars::fevd()` and Geweke decomposition tools partition predictive contribution by frequency band.

R Packages Used

`lmtest::grangertest()` for the bivariate test; `vars::VAR()` and `causality()` for VAR-based multivariate Granger causality; `tsDyn` for non-linear extensions; `NlinTS` for non-linear and information-theoretic causality measures.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics